# MANTA

*The ocean makes the oxygen we breathe, and we fill it with plastic.*

*MANTA is a wave from that ocean, made by an algorithm so you print only what you need.*




## THE IDEA

At least half of the oxygen we breathe comes from the ocean, so MANTA is a story of life: a wave that rose from the sea floor and stayed, one continuous ribbon printed at home in 43 parts and glued without screws. Its forked feet are the fins of a manta, the ray that tells us how healthy the sea is. From the side, legs, seat and back draw an M, for Manta and for *Mare*. Its colour is the sea itself, lightening into foam at the crest, and every seam is left white, like spray.

Yet we fill that ocean with plastic. Microplastics are now found even in human blood, and more and more of us, children too, print plastic at home. MANTA is plastic as well, so it is printed with sense: material only where the load is, almost nothing wasted, and recycled PETG recommended.

## PRINTED AT HOME

Every part fits a home printer bed. The algorithm turns each one to the position that needs the least support and shapes its holes for it, so supports take under 1 % of the filament and none is trapped inside a hole. In PETG, at 0.2 mm layers and 35 % gyroid, the chair weighs about 16.5 kg.

## ASSEMBLED LIKE FURNITURE

The ribbon is cut crosswise, away from the knee and lumbar where bending peaks, so every joint is a full solid cross-section glued face to face. A printed conical peg centres each one with 0.12 mm of play, tested on a printed sample; the two most loaded joints are also pinned, and where the crest returns through the backrest it is glued on as a saddle. The numbered parts go together in six steps, shown on the board.

## STRONG BY NATURE

Like bone, the ribbon is thickest where it bends most (Wolff's law). Like branching vessels, its lumbar balances three branches (Murray's law). The crest closes into a loop like a fern's fiddlehead, and each foot forks into two roots, a wide stance from little material. Checked at 120 kg with safety factors, its weakest glued joint holds 2.1 times the load; it resists tipping and passes all eight ergonomic checks. A full-size prototype is printed and standing; the photos are on the board.

## OPEN TO EVERYONE

MANTA is generated by an algorithm written for it. Choose who the chair is for and how thick it should be, and it rebuilds every cut, peg and joint, rechecks strength and stability, and refuses to export a chair that would fail. So anyone can print a lighter or stronger MANTA and use only the plastic they need: the 70 kg version built here takes half the filament. The source is open to download, change and print.

**Anonymised source: anonymous.4open.science/r/manta-chair-23D4**